In [13]:
```
import pandas as pd
```

In [15]:
```
import pandas as pd
```

In [17]:
```
import numpy as np
```

In [19]:
```
import matplotlib.pyplot as plt
```

In [21]:
```
wcat=pd.read_csv("C:\\Users\\asish\\OneDrive\\Desktop\\datascience\\Data Science
```

In [23]:
```
wcat
```

Out[23]:

	Waist	AT
0	74.75	25.72
1	72.60	25.89
2	81.80	42.60
3	83.95	42.80
4	74.65	29.84
...	...	...
104	100.10	124.00
105	93.30	62.20
106	101.80	133.00
107	107.90	208.00
108	108.50	208.00

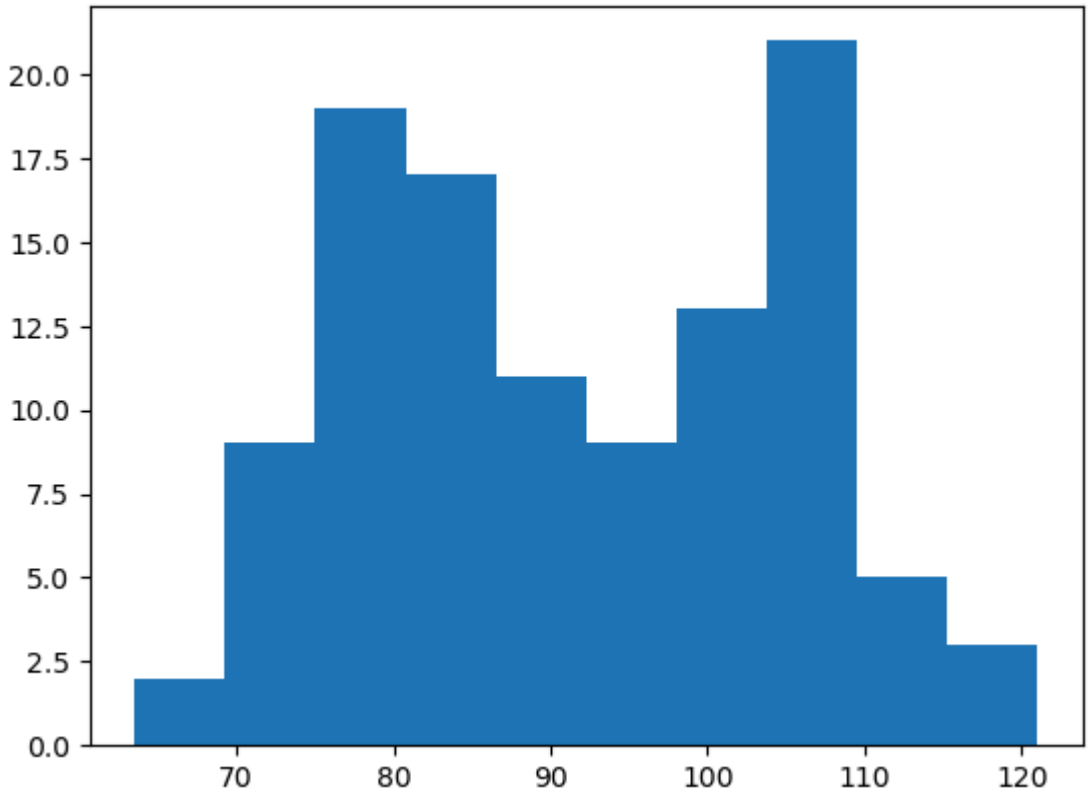
109 rows × 2 columns

In [25]:
```
wcat.columns
```

Out[25]:
```
Index(['Waist', 'AT'], dtype='object')
```

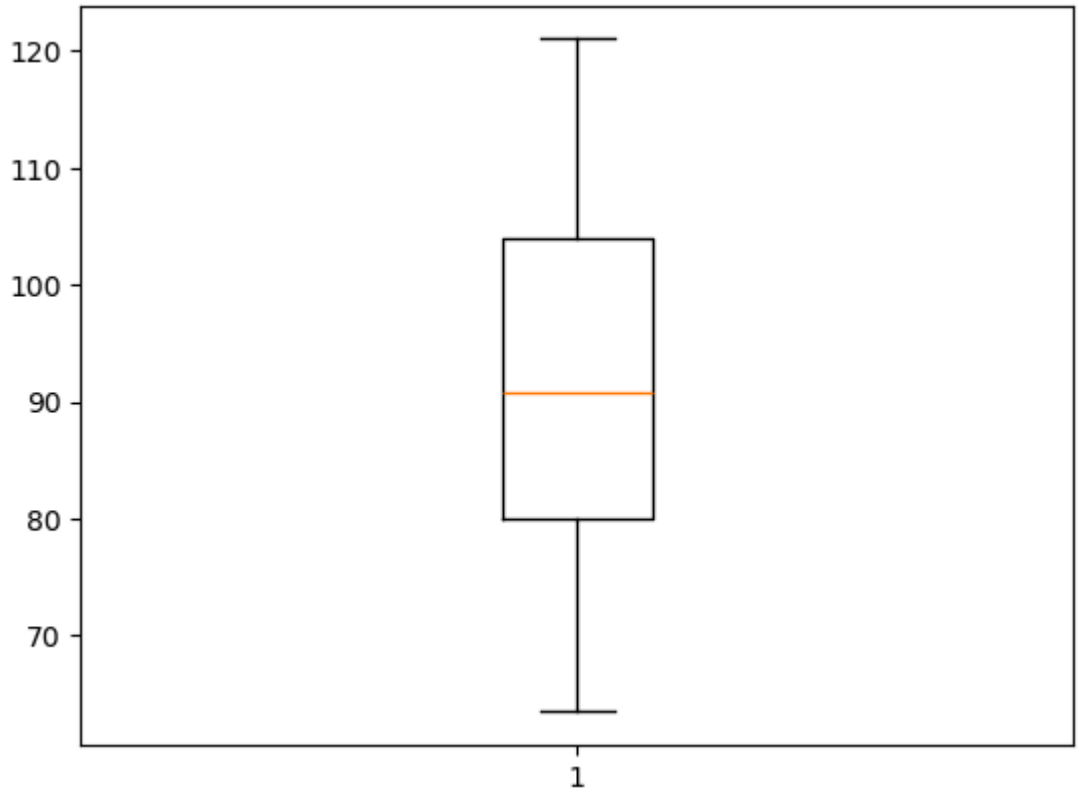
In [27]:
```
plt.hist(wcat.Waist)
```

Out[27]:
```
(array([ 2.,  9., 19., 17., 11.,  9., 13., 21.,  5.,  3.]),
 array([ 63.5 ,  69.25,  75.  ,  80.75,  86.5 ,  92.25,  98.  , 103.75,
        109.5 , 115.25, 121.  ]),
 <BarContainer object of 10 artists>)
```



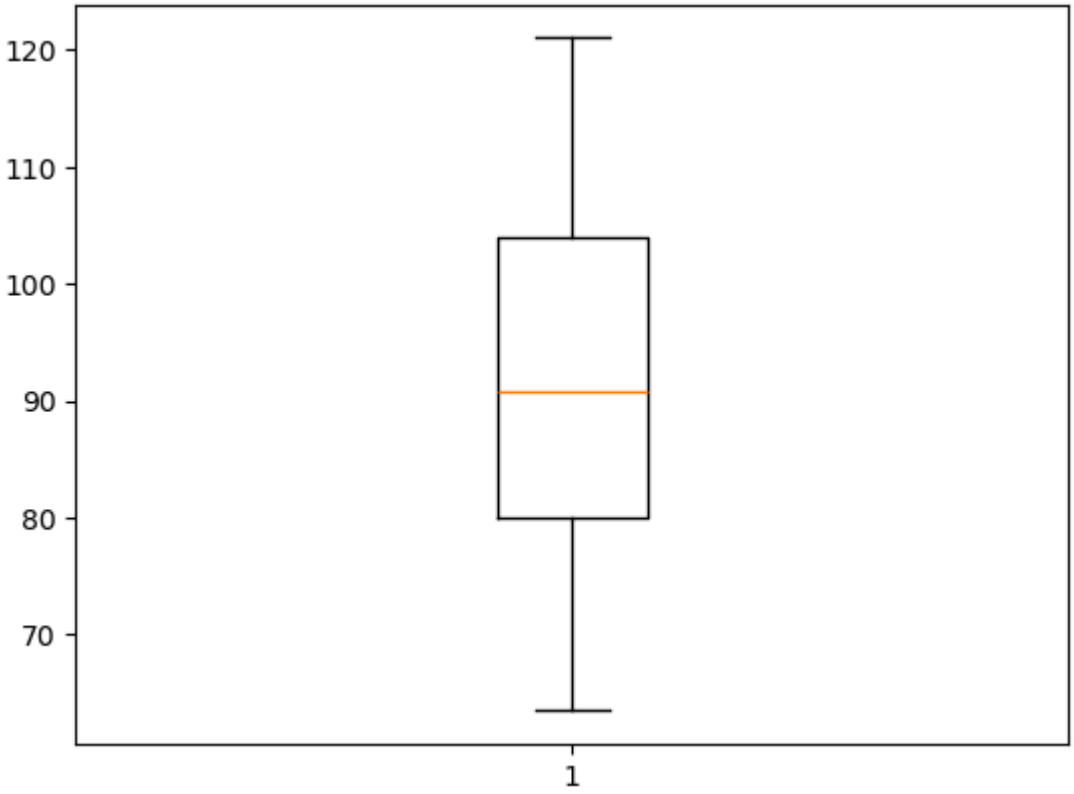
In [28]:
```
plt.boxplot(wcat.Waist)
```

Out[28]:
```
{'whiskers': [<matplotlib.lines.Line2D at 0x1f3f25223f0>,
  <matplotlib.lines.Line2D at 0x1f3f25d2930>],
 'caps': [<matplotlib.lines.Line2D at 0x1f3f25d2bd0>,
  <matplotlib.lines.Line2D at 0x1f3f25d2ea0>],
 'boxes': [<matplotlib.lines.Line2D at 0x1f3f2520710>],
 'medians': [<matplotlib.lines.Line2D at 0x1f3f25d3140>],
 'fliers': [<matplotlib.lines.Line2D at 0x1f3f25d3440>],
 'means': []}
```



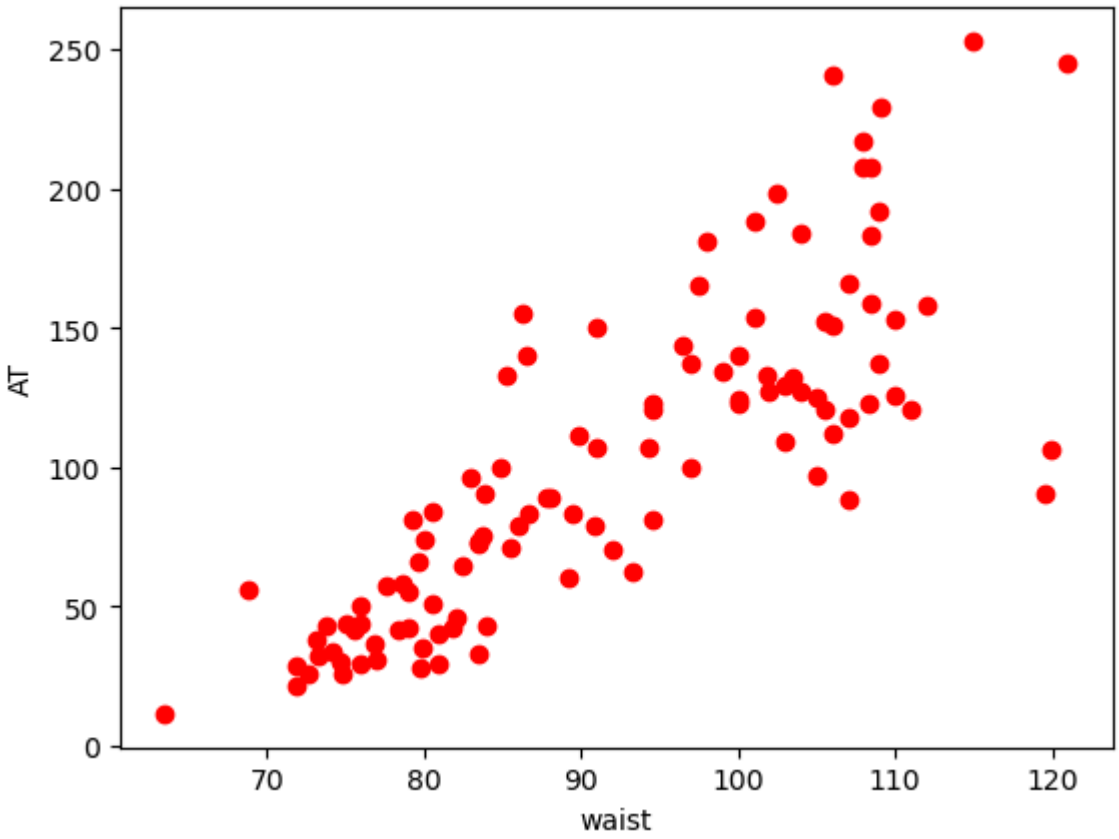
In [30]:
```
plt.boxplot(wcat.Waist,0,"rs",1)
```

Out[30]:
```
{'whiskers': [<matplotlib.lines.Line2D at 0x1f3f2e46030>,
  <matplotlib.lines.Line2D at 0x1f3f2e46330>],
 'caps': [<matplotlib.lines.Line2D at 0x1f3f2e465a0>,
  <matplotlib.lines.Line2D at 0x1f3f2e46750>],
 'boxes': [<matplotlib.lines.Line2D at 0x1f3f2df7dd0>],
 'medians': [<matplotlib.lines.Line2D at 0x1f3f2e46a20>],
 'fliers': [<matplotlib.lines.Line2D at 0x1f3f2e46c90>],
 'means': []}
```



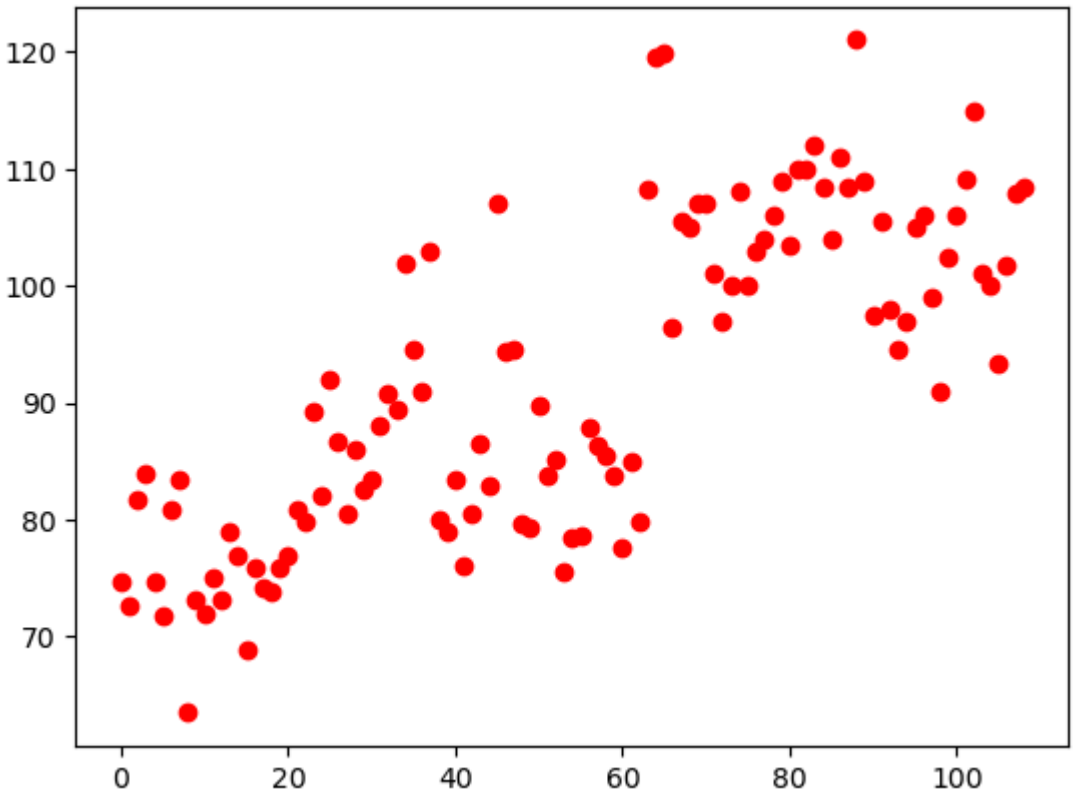
In [31]:
```
plt.plot(wcat.Waist,wcat.AT,"ro");plt.xlabel("waist");plt.ylabel("AT")
```

Out[31]:
```
Text(0, 0.5, 'AT')
```



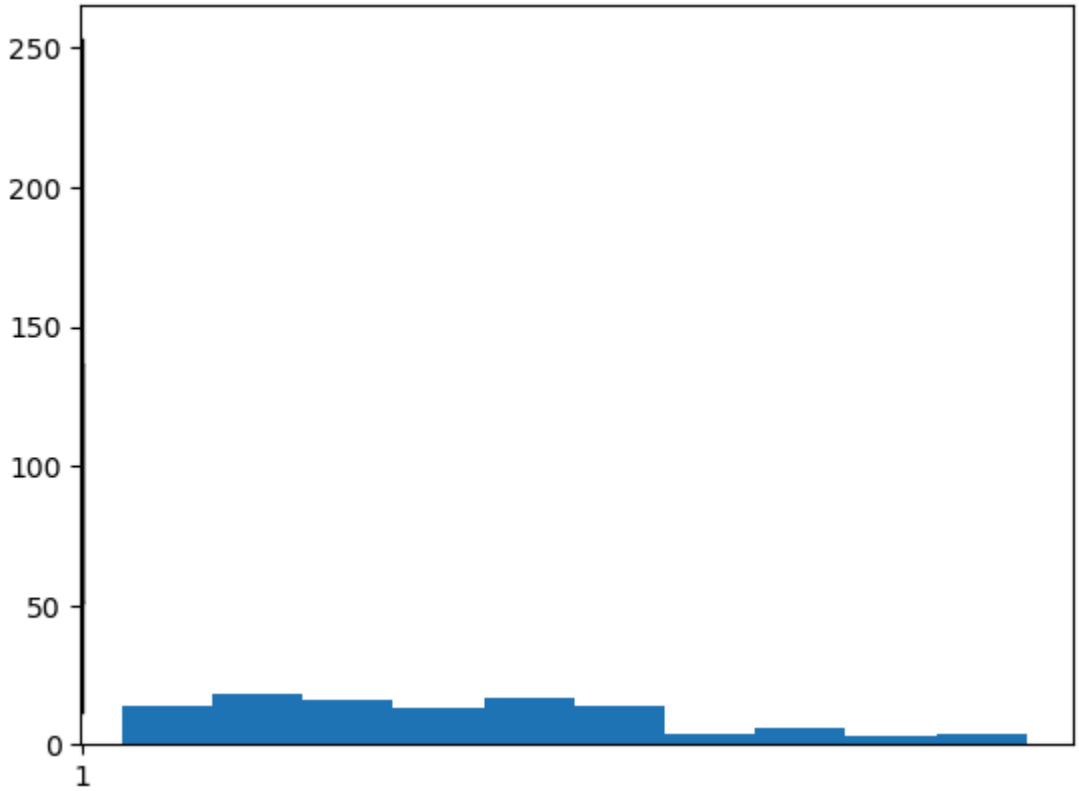
In [34]:
```
plt.plot(np.arange(109),wcat.Waist,"ro")
```

Out[34]:
```
[<matplotlib.lines.Line2D at 0x1f3f2efcce0>]
```



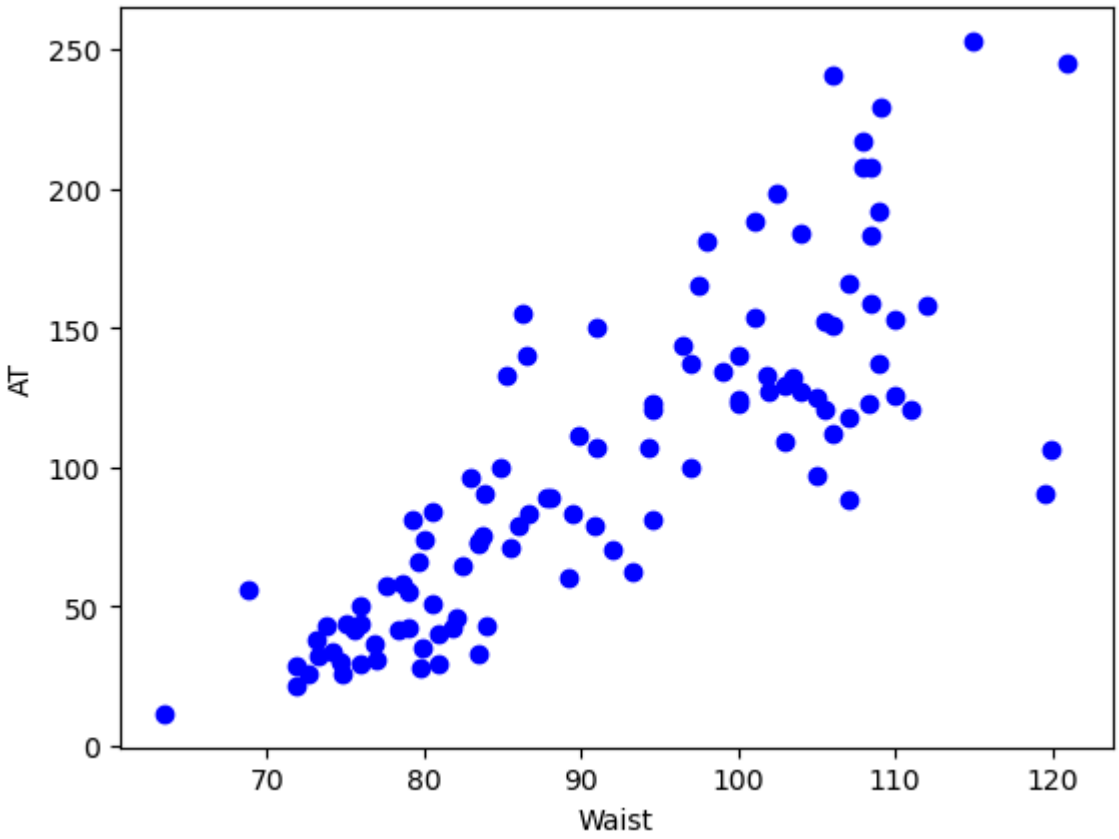
In [37]:
```
plt.hist(wcat.AT)
plt.boxplot(wcat.AT)
```

Out[37]:
```
{'whiskers': [<matplotlib.lines.Line2D at 0x1f3f2fa45f0>,
  <matplotlib.lines.Line2D at 0x1f3f2fa48c0>],
 'caps': [<matplotlib.lines.Line2D at 0x1f3f2fa49b0>,
  <matplotlib.lines.Line2D at 0x1f3f2fa4e60>],
 'boxes': [<matplotlib.lines.Line2D at 0x1f3f2fa4440>],
 'medians': [<matplotlib.lines.Line2D at 0x1f3f2fa5100>],
 'fliers': [<matplotlib.lines.Line2D at 0x1f3f2fa5400>],
 'means': []}
```



In [39]:
```
plt.plot(wcat.Waist,wcat.AT,"bo");plt.xlabel("Waist");plt.ylabel("AT")
```

Out[39]:
```
Text(0, 0.5, 'AT')
```



In [41]:
```
wcat.corr()
```

Out[41]:

	Waist	AT
Waist	1.000000	0.818558
AT	0.818558	1.000000

In [43]:
```
wcat.AT.corr(wcat.Waist)
```

Out[43]:
```
0.818557812895853
```

In [45]:
```
np.corrcoef(wcat.AT,wcat.Waist)
```

Out[45]:
```
array([[1.        , 0.81855781],
       [0.81855781, 1.        ]])
```

In [47]:
```
import statsmodels.formula.api as smf
```

In [48]:
```
model = smf.ols("AT~Waist",data=wcat).fit()
type(model)
```

Out[48]:
```
statsmodels.regression.linear_model.RegressionResultsWrapper
```

In [49]:
```
model.params
```

Out[49]:
```
Intercept   -215.981488
Waist          3.458859
dtype: float64
```

In [50]:
```
model.summary()
```

Out[50]:

OLS Regression Results

Dep. Variable:	AT	**R-squared:**	0.670
Model:	OLS	**Adj. R-squared:**	0.667
Method:	Least Squares	**F-statistic:**	217.3
Date:	Sun, 01 Mar 2026	**Prob (F-statistic):**	1.62e-27
Time:	17:45:38	**Log-Likelihood:**	-534.99
No. Observations:	109	**AIC:**	1074.
Df Residuals:	107	**BIC:**	1079.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
Intercept	-215.9815	21.796	-9.909	0.000	-259.190	-172.773
Waist	3.4589	0.235	14.740	0.000	2.994	3.924

Omnibus:	3.960	**Durbin-Watson:**	1.560
Prob(Omnibus):	0.138	**Jarque-Bera (JB):**	4.596
Skew:	0.104	**Prob(JB):**	0.100
Kurtosis:	3.984	**Cond. No.**	639.

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

In [51]:
```
model.conf_int(0.05)
```

Out[51]:

	0	**1**
Intercept	-259.190053	-172.772923
Waist	2.993689	3.924030

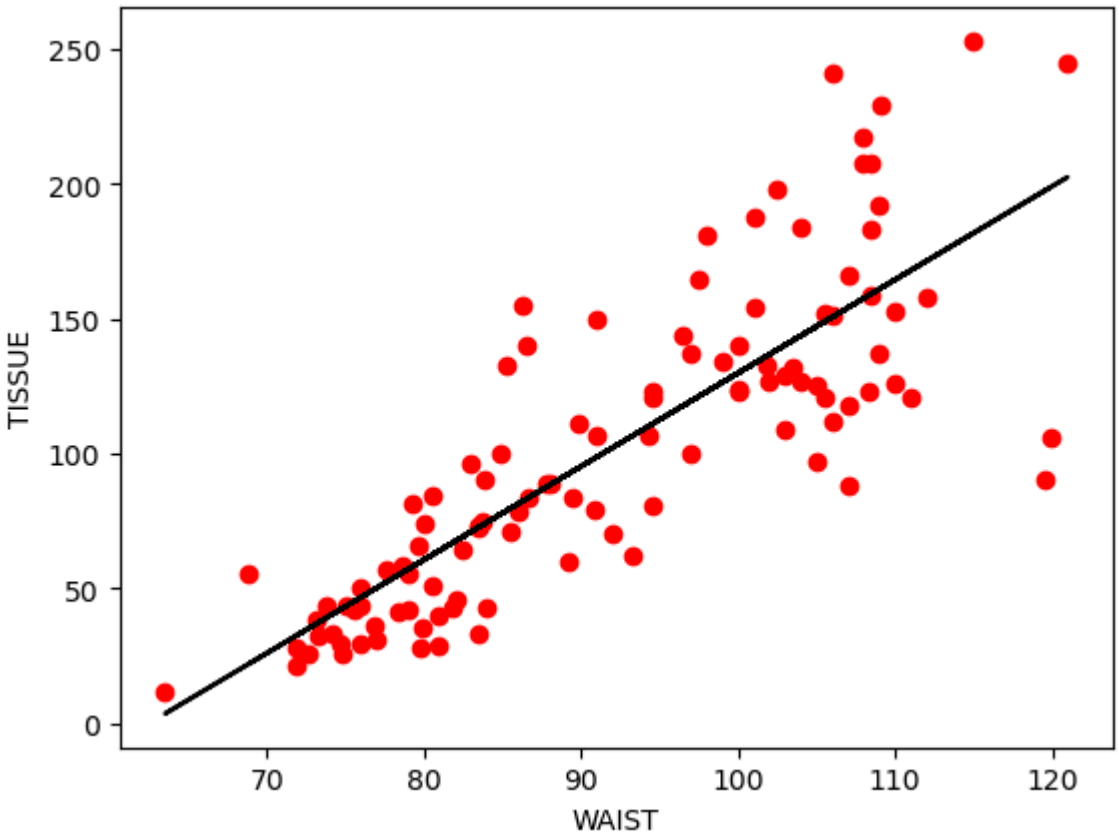
In [52]:
```
pred = model.predict(wcat)
```

In [53]:
```
pred
```

Out[53]:
```
0       42.568252
1       35.131704
2       66.953210
3       74.389758
4       42.222366
          ...
104    130.250337
105    106.730093
106    136.130398
107    157.229440
108    159.304756
Length: 109, dtype: float64
```

In [61]:
```
plt.scatter(x=wcat['Waist'],y=wcat['AT'],color='red');plt.plot(wcat['Waist'],pre
```

Out[61]:
```
Text(0, 0.5, 'TISSUE')
```



In [62]:
```
pred.corr(wcat.AT)
```

Out[62]:
```
0.8185578128958534
```

In [65]:
```
model2 = smf.ols('AT~np.log(Waist)',data=wcat).fit()
```

In [67]:
```
model2
```

Out[67]:
```
<statsmodels.regression.linear_model.RegressionResultsWrapper at 0x1f3f4a7e030>
```

In [69]:
```
model2.params
```

Out[69]:
```
Intercept       -1328.341989
np.log(Waist)     317.135564
dtype: float64
```

In [71]:
```
model2.summary()
```

Out[71]:

OLS Regression Results

Dep. Variable:	AT	**R-squared:**	0.675
Model:	OLS	**Adj. R-squared:**	0.672
Method:	Least Squares	**F-statistic:**	222.6
Date:	Sun, 01 Mar 2026	**Prob (F-statistic):**	6.80e-28
Time:	17:45:41	**Log-Likelihood:**	-534.11
No. Observations:	109	**AIC:**	1072.
Df Residuals:	107	**BIC:**	1078.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
Intercept	-1328.3420	95.923	-13.848	0.000	-1518.498	-1138.186
np.log(Waist)	317.1356	21.258	14.918	0.000	274.994	359.277

Omnibus:	3.317	**Durbin-Watson:**	1.599
Prob(Omnibus):	0.190	**Jarque-Bera (JB):**	2.908
Skew:	0.235	**Prob(JB):**	0.234
Kurtosis:	3.647	**Cond. No.**	145.

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

In [73]:
```
print(model2.conf_int(0.01))
```

```
                        0            1
Intercept     -1579.905632 -1076.778346
np.log(Waist)   261.384641   372.886487
```

In [75]:
```
pred2 = model2.predict(wcat)
```

In [77]:
```
pred2
```

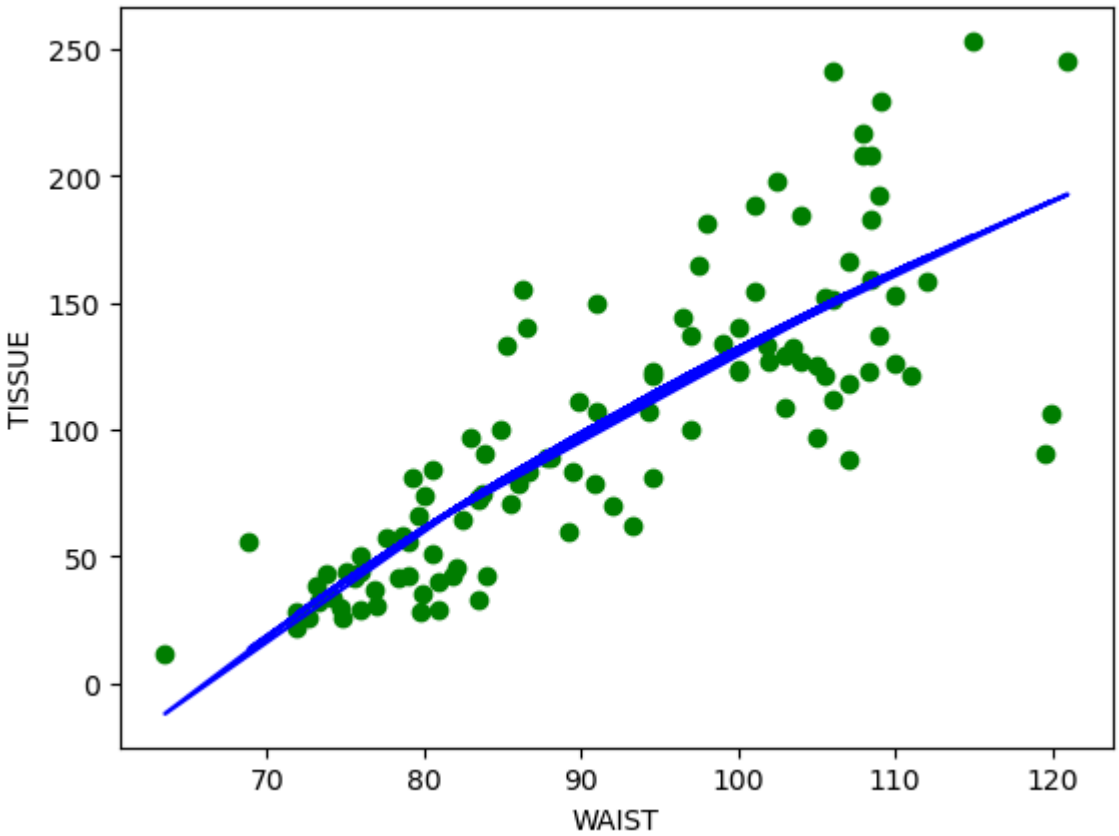
Out[77]:
```
0       39.828156
1       30.572779
2       68.410960
3       76.638769
4       39.403610
          ...
104    132.438233
105    110.127880
106    137.778929
107    156.234559
108    157.993171
Length: 109, dtype: float64
```

In [79]:
```
pred2.corr(wcat.AT)
```

Out[79]:
```
0.8217781862645354
```

In [81]:
```
plt.scatter(x=wcat['Waist'],y=wcat['AT'],color='green');plt.plot(wcat['Waist'],p
```

Out[81]:
```
Text(0, 0.5, 'TISSUE')
```



In [83]:
```
model3 = smf.ols('np.log(AT)~Waist',data=wcat).fit()
```

In [85]:
```
model3
```

Out[85]:
```
<statsmodels.regression.linear_model.RegressionResultsWrapper at 0x1f3f531f3e0>
```

In [87]:
```
model3.params
```

Out[87]:
```
Intercept    0.741021
Waist        0.040252
dtype: float64
```

In [89]:
```
model3.summary()
```

Out[89]:

OLS Regression Results

Dep. Variable:	np.log(AT)	**R-squared:**	0.707
Model:	OLS	**Adj. R-squared:**	0.704
Method:	Least Squares	**F-statistic:**	258.3
Date:	Sun, 01 Mar 2026	**Prob (F-statistic):**	2.68e-30
Time:	17:45:45	**Log-Likelihood:**	-40.123
No. Observations:	109	**AIC:**	84.25
Df Residuals:	107	**BIC:**	89.63
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	0.7410	0.233	3.185	0.002	0.280	1.202
Waist	0.0403	0.003	16.073	0.000	0.035	0.045

Omnibus:	2.604	**Durbin-Watson:**	1.201
Prob(Omnibus):	0.272	**Jarque-Bera (JB):**	2.117
Skew:	-0.329	**Prob(JB):**	0.347
Kurtosis:	3.182	**Cond. No.**	639.

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

In [91]:
```
print(model3.conf_int(0.01))
```

```
                  0         1
Intercept  0.130939  1.351102
Waist      0.033684  0.046820
```

In [93]:
```
pred_log = model3.predict(wcat)
```

In [95]:
```
pred_log
```

Out[95]:
```
0      3.749889
1      3.663346
2      4.033669
3      4.120211
4      3.745864
         ...
104    4.770288
105    4.496571
106    4.838717
107    5.084257
108    5.108408
Length: 109, dtype: float64
```

In [97]:
```
pred3=np.exp(pred_log)
```

In [99]:
```
pred3
```

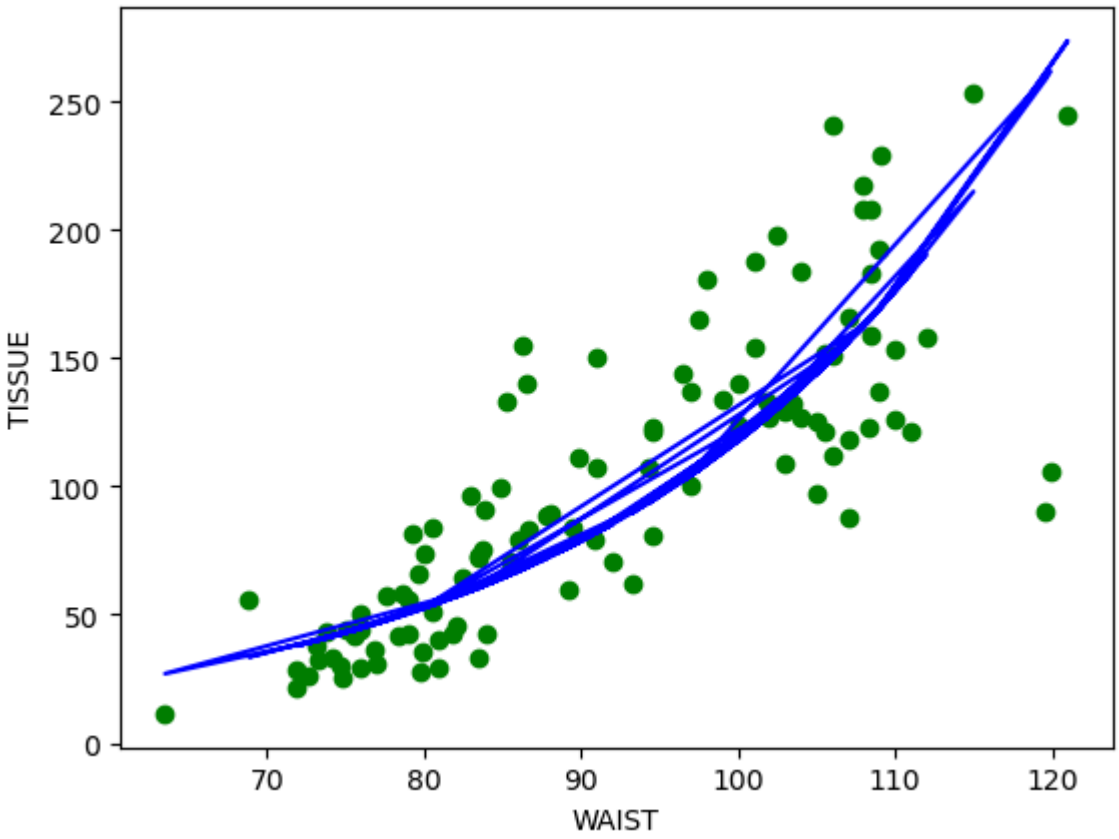
Out[99]:
```
0       42.516361
1       38.991601
2       56.467684
3       61.572246
4       42.345566
          ...
104    117.953181
105     89.709020
106    126.307181
107    161.459872
108    165.406833
Length: 109, dtype: float64
```

In [101…
```
pred3.corr(wcat.AT)
```

Out[101…
```
0.7633804583650526
```

In [103…
```
plt.scatter(x=wcat['Waist'],y=wcat['AT'],color='green');plt.plot(wcat.Waist,np.e
```

Out[103…
```
Text(0, 0.5, 'TISSUE')
```



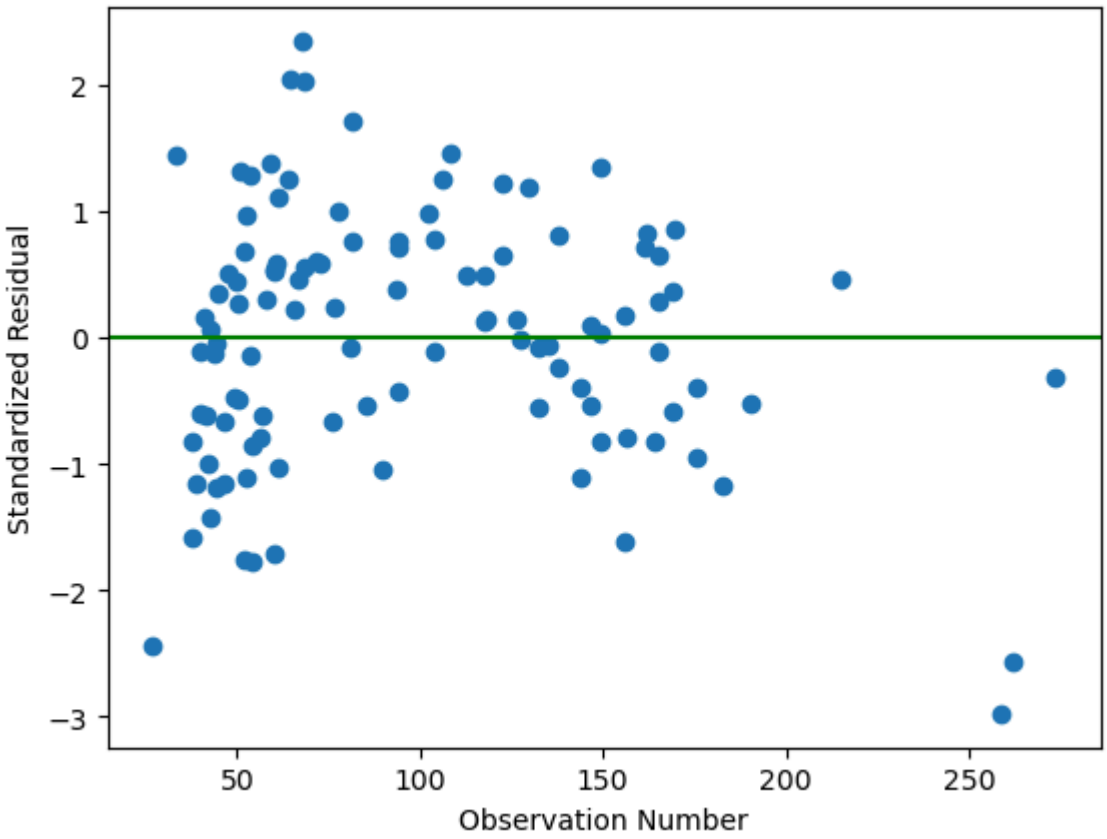
In [105…
```
resid_3 = pred3-wcat.AT
```

In [107…
```
resid_3
```

Out[107…
```
0      16.796361
1      13.101601
2      13.867684
3      18.772246
4      12.505566
          ...
104    -6.046819
105    27.509020
106    -6.692819
107   -46.540128
108   -42.593167
Length: 109, dtype: float64
```

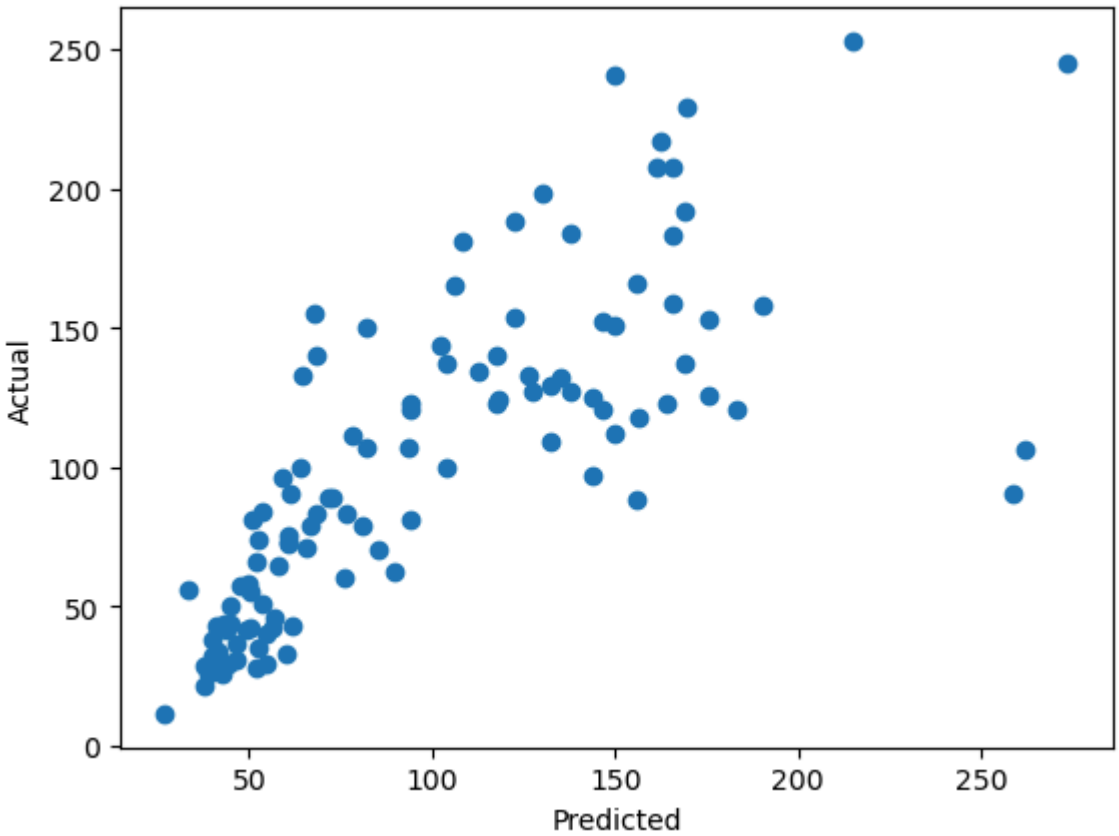
In [109…
```
plt.plot(pred3,model3.resid_pearson,"o");plt.axhline(y=0,color='green');plt.xlab
```

Out[109…
```
Text(0, 0.5, 'Standardized Residual')
```



In [111…
```
plt.scatter(x=pred3,y=wcat.AT);plt.xlabel("Predicted");plt.ylabel("Actual")
```

Out[111…
```
Text(0, 0.5, 'Actual')
```



In [113…
```
wcat["Waist_Sq"] = wcat.Waist*wcat.Waist
```

In [115…
```
model_quad = smf.ols("AT~Waist+Waist_Sq",data=wcat).fit()
```

In [117…
```
model_quad.params
```

Out[117…
```
Intercept   -466.288707
Waist          8.970879
Waist_Sq      -0.029700
dtype: float64
```

In [119…
```
model_quad.summary()
```

Out[119…

OLS Regression Results

Dep. Variable:	AT	**R-squared:**	0.678
Model:	OLS	**Adj. R-squared:**	0.672
Method:	Least Squares	**F-statistic:**	111.6
Date:	Sun, 01 Mar 2026	**Prob (F-statistic):**	8.37e-27
Time:	17:45:53	**Log-Likelihood:**	-533.67
No. Observations:	109	**AIC:**	1073.
Df Residuals:	106	**BIC:**	1081.
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	-466.2887	156.964	-2.971	0.004	-777.485	-155.092
Waist	8.9709	3.431	2.614	0.010	2.168	15.774
Waist_Sq	-0.0297	0.018	-1.610	0.110	-0.066	0.007

Omnibus:	3.682	**Durbin-Watson:**	1.620
Prob(Omnibus):	0.159	**Jarque-Bera (JB):**	3.054
Skew:	0.338	**Prob(JB):**	0.217
Kurtosis:	3.465	**Cond. No.**	4.49e+05

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 4.49e+05. This might indicate that there are
strong multicollinearity or other numerical problems.

In [121…
```
pred_quad = model_quad.predict(wcat)
```

In [123…
```
model_quad.conf_int(0.05)
```

Out[123…

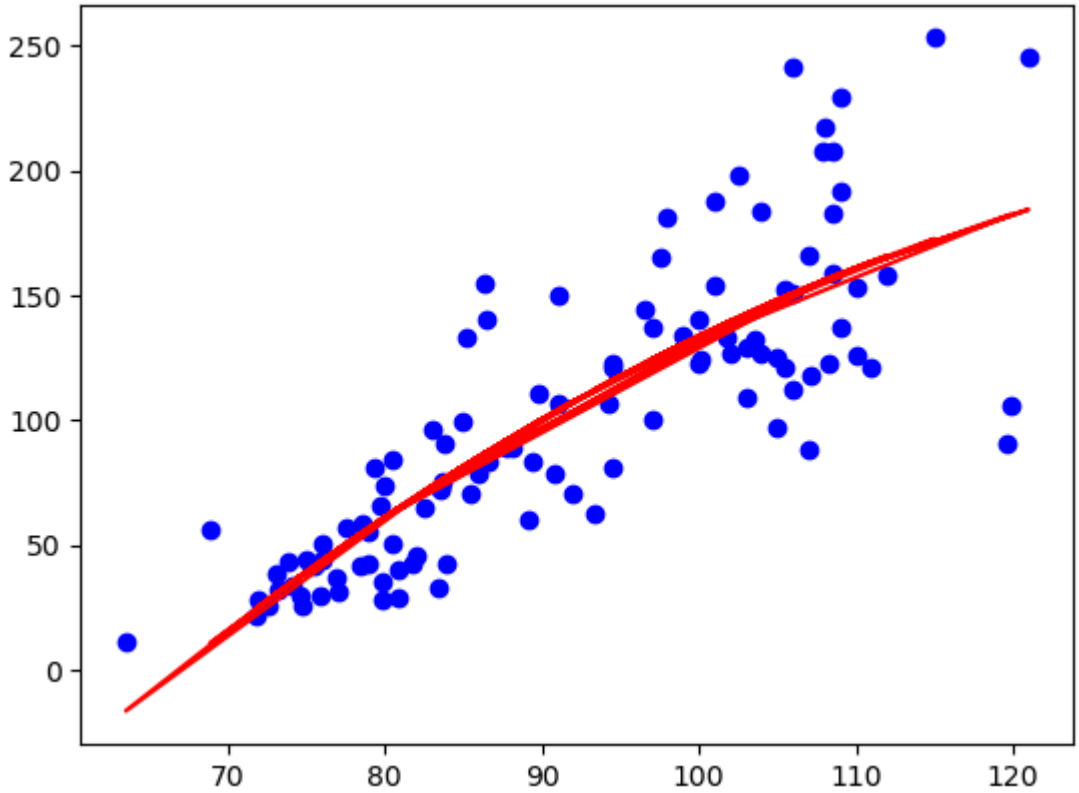
	0	1
Intercept	-777.484955	-155.092458
Waist	2.167743	15.774014
Waist_Sq	-0.066273	0.006872

In [125…

```
plt.scatter(wcat.Waist,wcat.AT,c="b");plt.plot(wcat.Waist,pred_quad,"r")
```

Out[125…

```
[<matplotlib.lines.Line2D at 0x1f3f64123c0>]
```

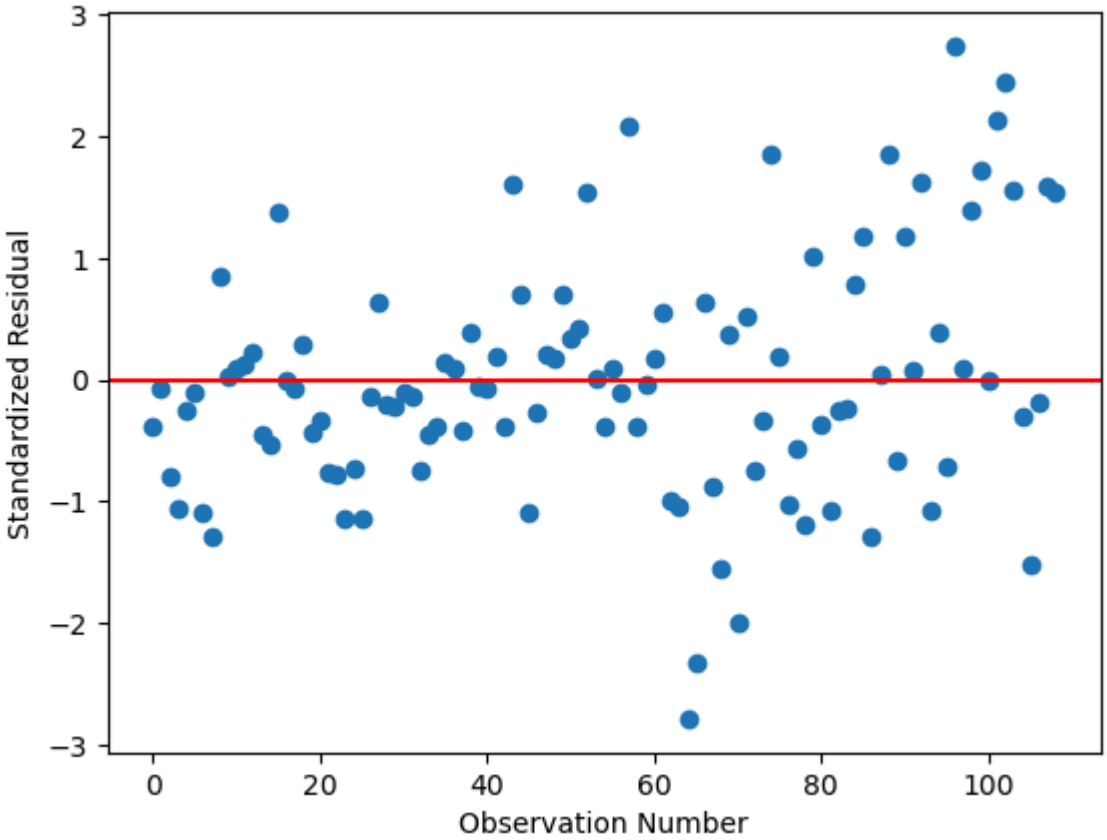


In [127…

```
plt.scatter(np.arange(109),model_quad.resid_pearson);plt.axhline(y=0,color='red'
```

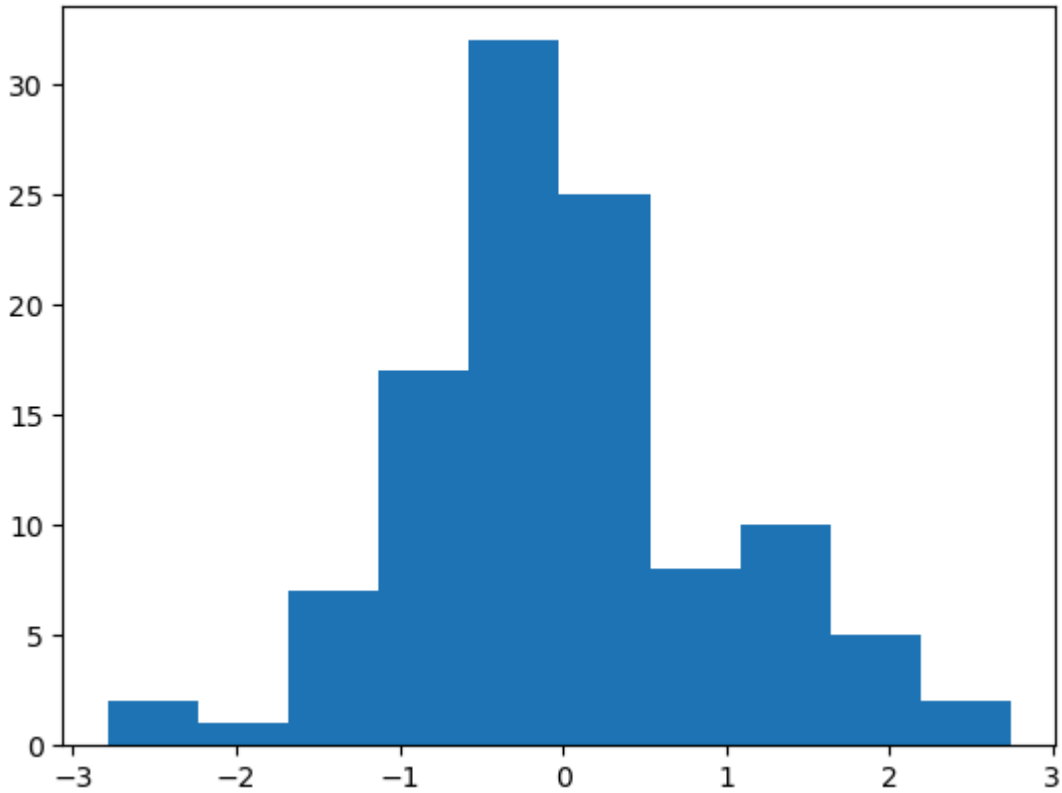
Out[127…

```
Text(0, 0.5, 'Standardized Residual')
```



In [129…
```
plt.hist(model_quad.resid_pearson)
```

Out[129…
```
(array([ 2.,  1.,  7., 17., 32., 25.,  8., 10.,  5.,  2.]),
 array([-2.78415636, -2.2312627 , -1.67836904, -1.12547538, -0.57258173,
        -0.01968807,  0.53320559,  1.08609925,  1.6389929 ,  2.19188656,
         2.74478022]),
 <BarContainer object of 10 artists>)
```

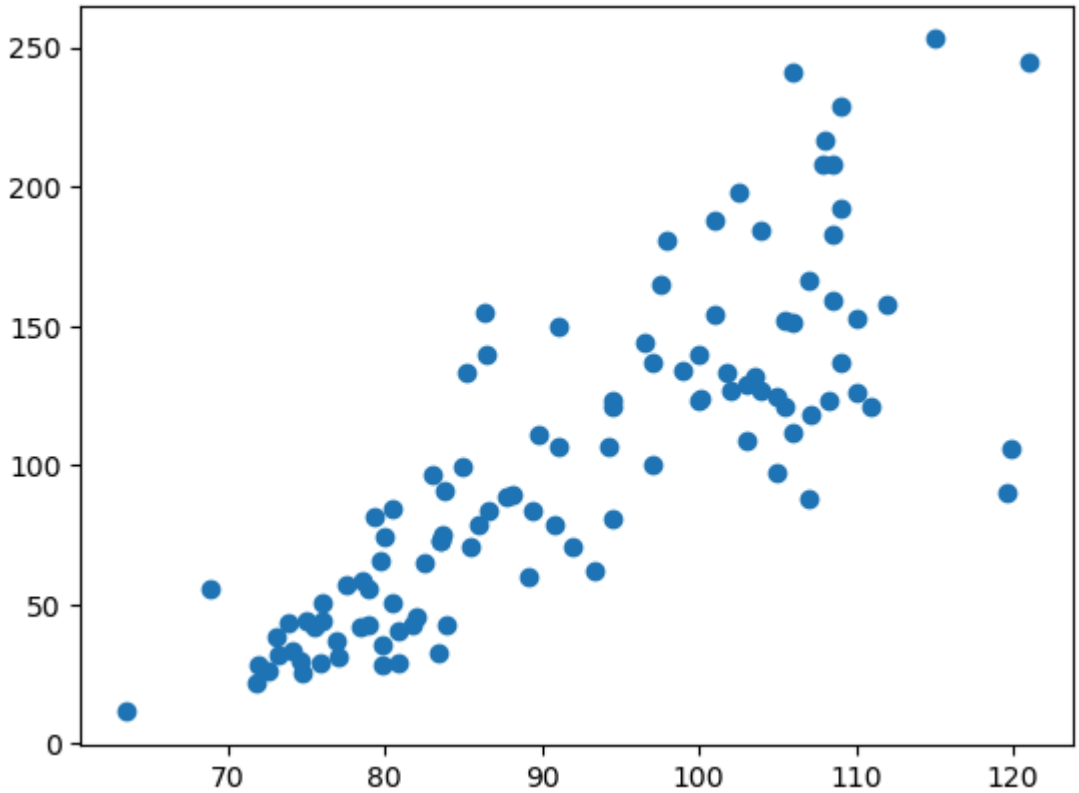


In [131…
```
from sklearn.linear_model import LinearRegression
```

In [132…
```
import numpy as np
```

In [135…
```
plt.scatter(wcat.Waist,wcat.AT)
```

Out[135…
```
<matplotlib.collections.PathCollection at 0x1f3f6e248f0>
```



In [137…
```
model1 = LinearRegression()
```

In [139…
```
model1
```

Out[139…
```
LinearRegression
LinearRegression()
```

In [141…
```
model1.fit(wcat.Waist.values.reshape(-1,1),wcat.AT)
```

Out[141…
```
LinearRegression
LinearRegression()
```

In [143…
```
pred1 = model1.predict(wcat.Waist.values.reshape(-1,1))
```

In [145…
```
pred1
```

Out[145…
```
array([ 42.56825159,  35.1317039 ,  66.95321031,  74.389758  ,
        42.22236565,  32.53755936,  63.84023685,  72.48738533,
         3.65608343,  37.20701954,  32.71050233,  43.43296644,
        36.8611336 ,  57.26840401,  50.35068522,  22.16098118,
        46.71888286,  40.49293596,  39.28233517,  46.54593989,
        49.83185632,  63.84023685,  60.38137746,  92.54876981,
        67.64498219, 102.23357611,  83.55573539,  62.4566931 ,
        81.48041975,  69.37441188,  72.83327127,  88.74402448,
        98.08294484,  93.24054169, 136.82217003, 110.88072459,
        98.77471671, 140.28102942,  60.7272634 ,  57.26840401,
        72.83327127,  46.89182583,  62.4566931 ,  83.20984945,
        71.10384158, 154.46235293, 110.18895271, 110.88072459,
        59.68960558,  58.30606183,  94.62408544,  73.87092909,
        78.71333224,  45.16239614,  55.19308837,  55.88486025,
        87.70636666,  82.51807757,  79.75099006,  73.52504315,
        52.42600086,  77.67567442,  60.03549152, 158.6129842 ,
       197.69809533, 198.73575314, 117.79844337, 148.9281779 ,
       147.1987482 , 154.11646699, 154.11646699, 133.36331063,
       119.52787307, 129.90445124, 157.57532638, 129.90445124,
       140.28102942, 143.73988881, 150.65760759, 161.03418577,
       142.01045911, 164.49304516, 164.49304516, 171.41076395,
       159.30475607, 143.73988881, 167.95190455, 159.30475607,
       202.54049847, 161.03418577, 121.25730276, 148.9281779 ,
       122.98673246, 110.88072459, 119.52787307, 147.1987482 ,
       150.65760759, 126.44559185,  98.77471671, 138.55159972,
       150.65760759, 161.38007171, 181.78734212, 133.36331063,
       130.25033718, 106.73009332, 136.13039815, 157.22944044,
       159.30475607])
```

In [147…
```
model1.score(wcat.Waist.values.reshape(-1,1),wcat.AT)# 0.6700
```

Out[147…
```
0.6700368930528429
```

In [149…
```
rmse1 = np.sqrt(np.mean((pred1-wcat.AT)**2))
```

In [151…
```
rmse1
```

Out[151…
```
32.760177495755144
```

In [153…
```
model1.coef_
```

Out[153…
```
array([3.45885939])
```

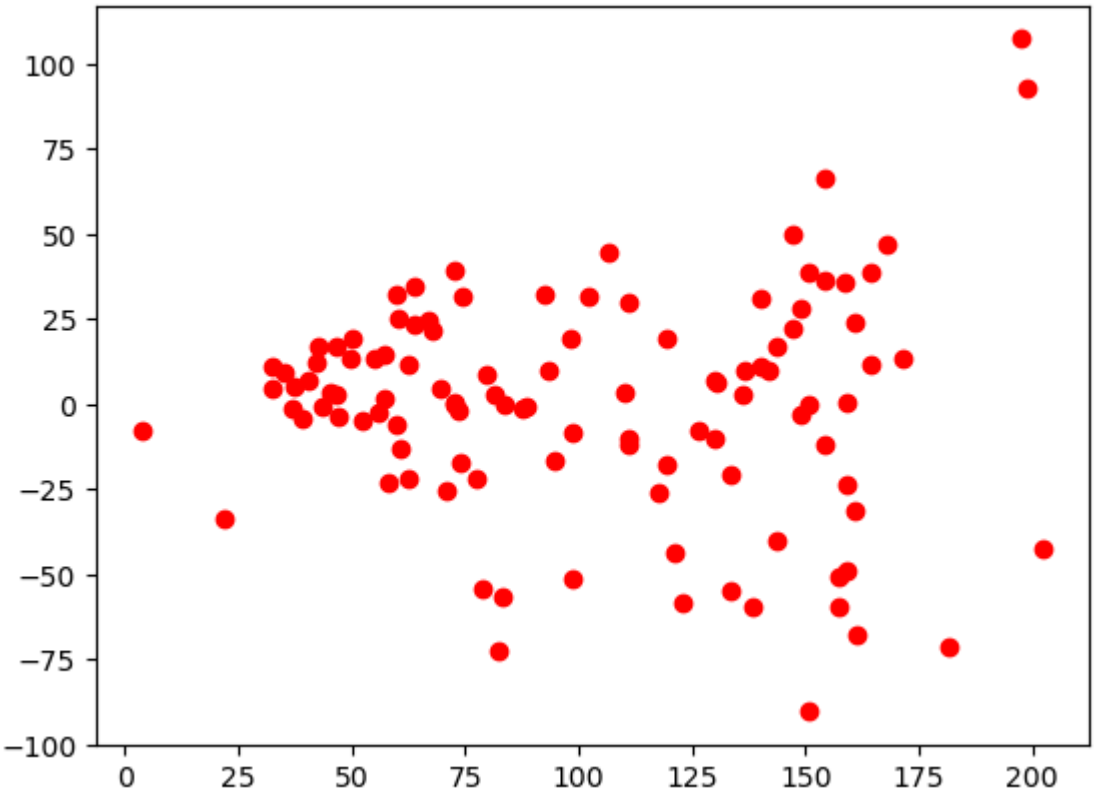
In [155…
```
model1.intercept_
```

Out[155…
```
-215.98148796164242
```

In [157…
```
import matplotlib.pyplot as plt
```

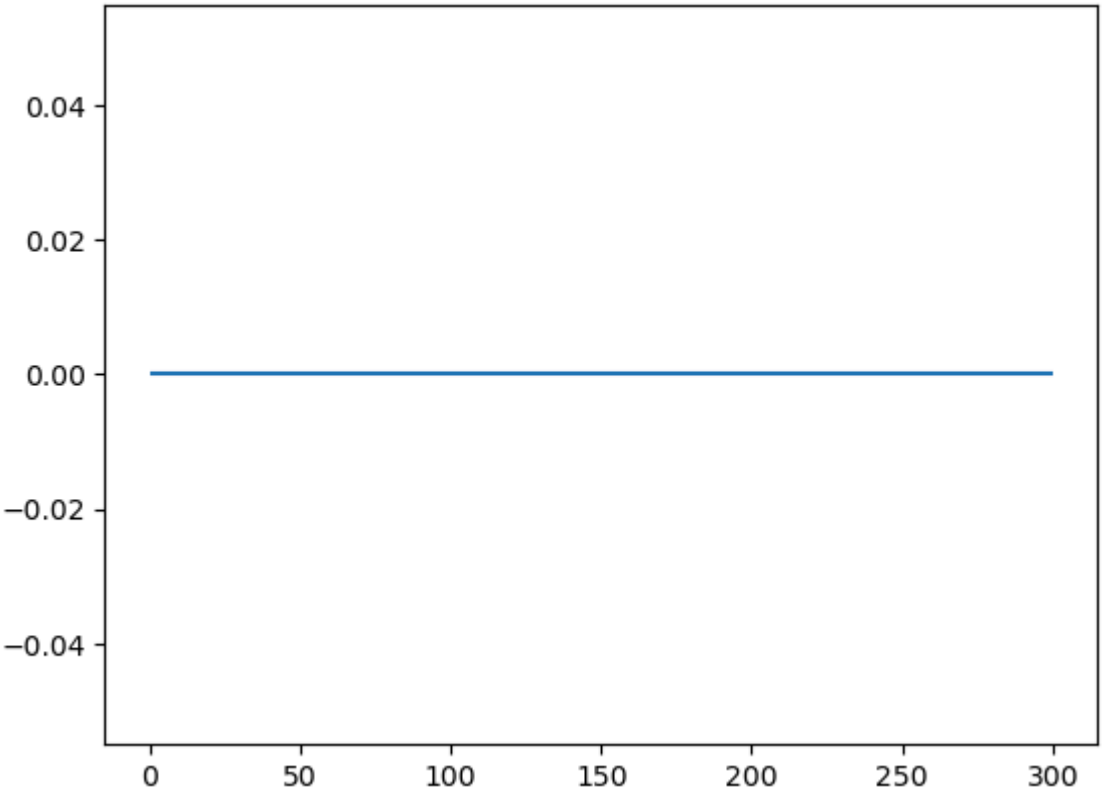
In [159…
```
plt.scatter(pred1,(pred1-wcat.AT),c="r")
```

Out[159…
```
<matplotlib.collections.PathCollection at 0x1f3f6ea4110>
```



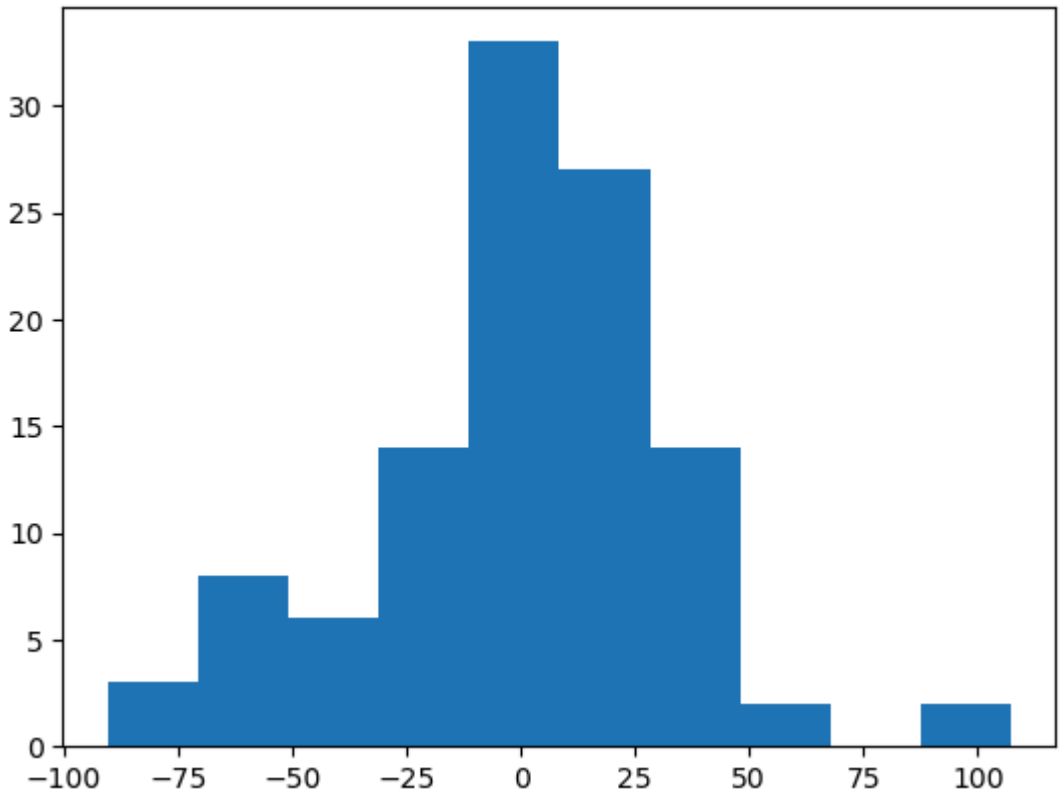
In [160…
```
plt.hlines(y=0,xmin=0,xmax=300)
```

Out[160…
```
<matplotlib.collections.LineCollection at 0x1f3f6d35be0>
```



In [163…
```
plt.hist(pred1-wcat.AT)
```

Out[163…
```
(array([ 3.,  8.,  6., 14., 33., 27., 14.,  2.,  0.,  2.]),
 array([-90.34239241, -70.57934363, -50.81629486, -31.05324609,
        -11.29019731,   8.47285146,  28.23590023,  47.99894901,
         67.76199778,  87.52504655, 107.28809533]),
 <BarContainer object of 10 artists>)
```



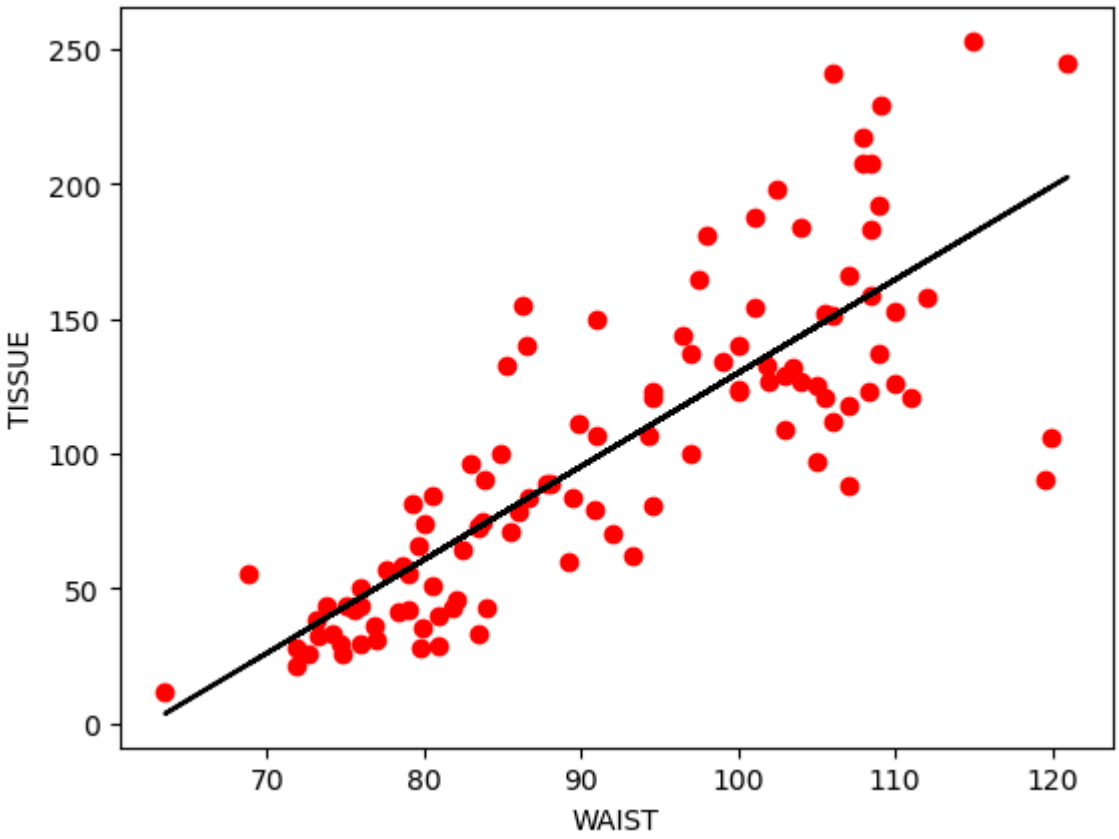
In [165…
```
from sklearn.ensemble import IsolationForest
```

In [167…
```
np.corrcoef(wcat.AT,wcat.Waist)
```

Out[167…
```
array([[1.        , 0.81855781],
       [0.81855781, 1.        ]])
```

In [169…
```
plt.scatter(x=wcat['Waist'],y=wcat['AT'],color='red');plt.plot(wcat['Waist'],pre
```

Out[169…
```
Text(0, 0.5, 'TISSUE')
```



In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]: